

Multimodal Ensembling

- for -

Zero-Shot Image Classification

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INTRODUCTION

THE ISSUE

Image classification is still challenging because **classes are inherently multimodal**.

Traditional approaches that provide solely images or descriptions of a given class, do not fully encapsulate the entire concept

HYPOTHESIS

Will ensembling with multiple **modalities** allow for more robust and generalizable image classification?

CHALLENGES

1. Models with low capacity often suffer from underfitting and thus **underperform** on fine-grained image classification.
2. It is essential to ensure high-quality data with rich **cross-modal representations** of each class, which often can be challenging to generate.

RESEARCH

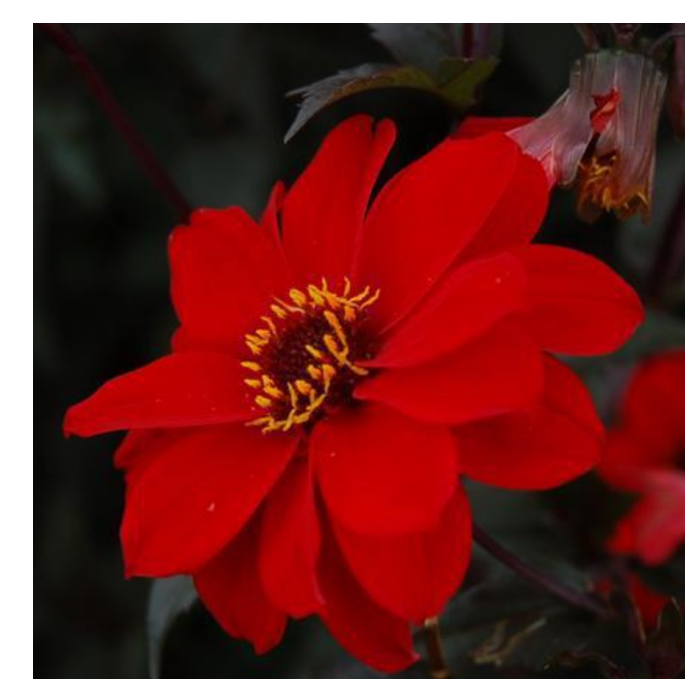
METHODOLOGY

Ensemble Learning forms a weighted sum of the embeddings produced by the classifiers. The ensemble consists of:

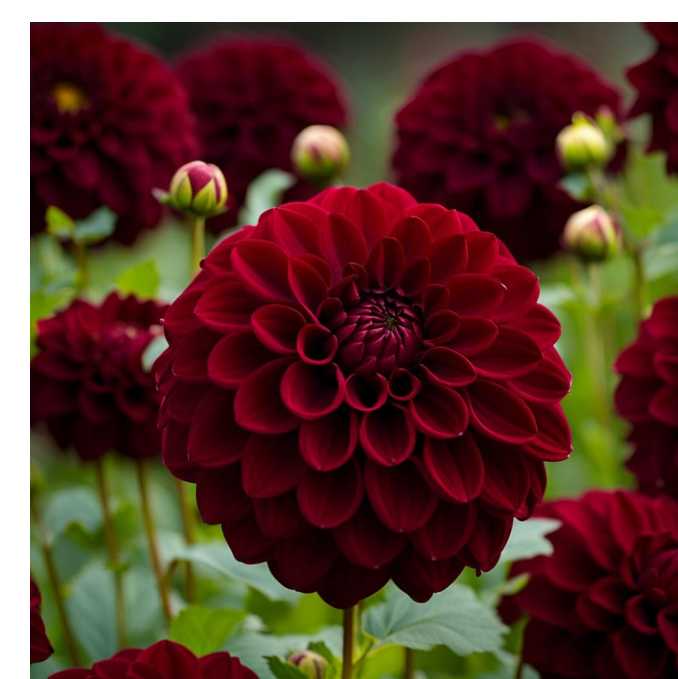
1. An **image-to-text** classifier that utilizes generated class descriptions
2. An **image-to-image** classifier that uses generated images for prediction.

ANALYSIS

The baseline is the standard method of top-1 image classification accuracy with the CLIP ViT-L/14 model.



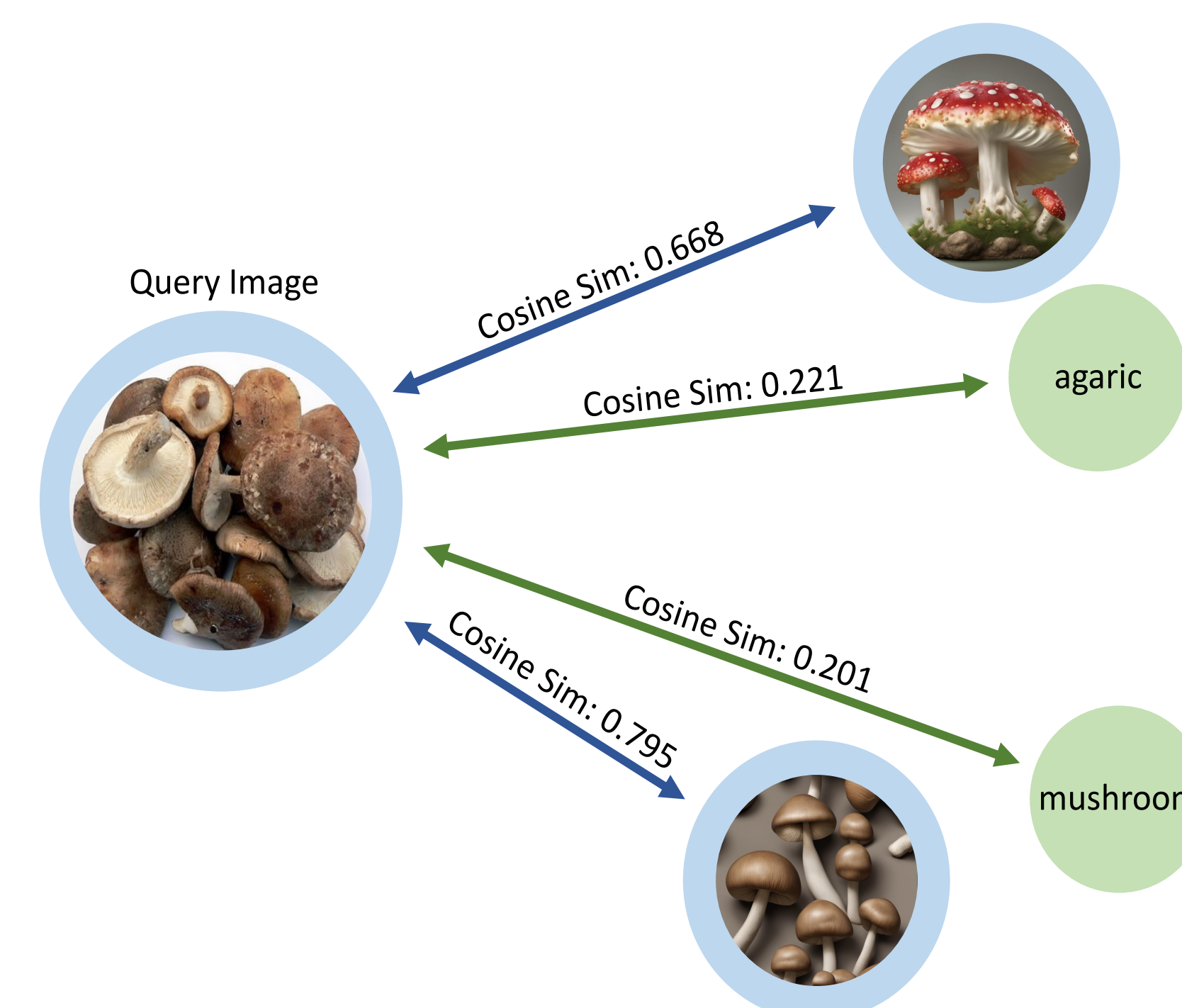
Bishop of Llandaff



"Dahlia 'Bishop of Llandaff' is a cultivar known for its dark red or maroon flowers and dark foliage."

CLIP Accuracy: 0%

Ensemble Accuracy: 45%



RESULTS

Preliminary results show significant improvements to top-1 image classification accuracy when compare to the CLIP ViT-L/14 baseline

	Flowers 102	DTD
CLIP ViT-L/14	74.43	51.54
Base Ensemble	77.98	52.02
CLIP ViT-L/14 Average Templates	78.81	55.18
Average Ensemble	80.51	55.46

CONCLUSION

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These results show that this improved conceptual representation from the multimodal approach directly results in **higher classification accuracy**.

DISCUSSION

Two prominent assumptions are made for this project:

- The generative models better represent the **true distribution** of the data (due to increased complexity and data diversity)
- The multimodal model can **distinguish** between multiple similar classes

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